Supplementary Notes to "How Many Atoms, Where, and When"

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These notes are submitted with the paper, whose three problems are the capacity, the least favorable prior (LFP) and the nonparametric maximum-likelihood estimate (NPMLE). They hold the side results that do not carry its argument. A prediction is registered when its value and an acceptance band were recorded in a dated log before the optimum was computed or read; the other terms are those of the paper. The first sections give optimal values and center events in d dimensions, the continuation of the planar rule past a birth and the rings inside rings. The next sections give the residual of the births law and the structure of its coefficients, and what does not carry over, with the full list of registered tests that did not pass. The last section proves the deformation identity of the NPMLE. Sections, tables, figures, equations and propositions of these notes carry the letter S (Section S1, Table S1, Fig. S1, equation (S1), Proposition S1); numbers without the letter S refer to the paper. Entry numbers in the tables are those of the public registration log. The references of these notes are numbered separately from those of the paper. Each section opens with a status line that says what is proved, what is checked numerically and what is conjectured.

S1. Optimal values in d dimensions: the widened region

Status. The capacity law (S1), the NPMLE perimeter law and the LFP law (S2) are numerical laws, and Table S1 lists the tests that support them or fail them. The inequality $d - B \le 4\lambda_1(\Omega)$ is proved in this section, and the deformation identity behind the NPMLE law is proved in Section S8 for the unconstrained problem.

A. Three value laws

The edge constants of Table I of the paper fix the optimal values on regions in two and more dimensions. Write Ω_w for Ω widened by distance w; for a convex planar region with perimeter P, Steiner's formula gives $|\Omega_w| = |\Omega| + wP + \pi w^2$.

Capacity. On a convex planar region with a smooth boundary whose radius of curvature is large compared with the strip,

$$e^{C+h_2} = |\Omega_{\beta/2}| + \pi(1 + 2k_r) + c \oint \kappa^2 \, ds + \cdots , \qquad \text{(S1)}$$

where κ is the curvature and k_r is the capacity reward response of Table I of the paper. The constant term is $\pi(1 + 2k_r) = -0.831$, and $c = -0.279$ is fitted on discs and not tested independently. The superellipse test resolves the shape term only to a few per cent (Table S1). For smooth regions whose reach exceeds $\beta/2$ the constant term becomes $(1 + 2k_r)\pi\chi(\Omega)$, with χ the Euler characteristic, which is zero for an annulus. On the d-ball of radius A we have $e^{C+h_d} = V_d A^d + S_d A^{d-1}\big(\beta/2 + a_d(d-1)/A + \cdots\big)$ with $a_d = \frac{1}{2} + (\beta/2)^2/2 + k_r = 1.84939$, where V_d and S_d are the volume and surface area of the unit ball. Of these coefficients, β is measured on certified one-dimensional states, and Section III proves that the lower and upper limits of $e^{C+h_1} - 2A$ lie in $(3.97966, 3.99001]$, k_r is fitted in one dimension (Table I of the paper), and c is fitted on discs.

NPMLE. For data uniform on a planar region, $G = D_N P + D'(0) \oint \kappa \, ds + b \oint \kappa^2 \, ds + c_3 \oint \kappa^3 \, ds + \cdots$, where D_N is measured in one dimension and $D'(0)$ is a post hoc value from one dimension (Table I of the paper). The coefficients b and c_3 are disc fits that depend on the truncation ($b = -0.0180$ or -0.0197 and $c_3 = -0.0803$ or -0.0456 without or with a κ^4 term). This is a perimeter law, and we name no widening width for it. In three dimensions each sphere of a concentric shell carries the ball's surface terms, including the term of order one. With H the mean curvature and κ_G the Gauss curvature, the integrals $\oint H^2 \, dA$ and $\oint \kappa_G \, dA$ both equal 4π on every sphere, so spheres cannot separate them, and other surfaces remain untested. Behind the planar law is an exact identity for the derivative of G under a deformation of the boundary, which is proved in Section S8 for the unconstrained problem; for convex regions this is the problem of the paper.

LFP. By the identity $B = d - I(p_\pi)$, maximizing the Bayes risk is minimizing the Fisher information of the output density $p_\pi = \pi * \varphi_d$ over priors π on Ω. The output density is positive everywhere, but smoothing does not increase Fisher information (by convexity of $(p, \nabla p) \mapsto |\nabla p|^2/p$), so $I(p_\pi) \le I(\pi)$. The least Fisher information of a density supported in Ω is $4\lambda_1(\Omega)$, with λ_1 the principal Dirichlet eigenvalue, since $I(\pi) = 4 \int |\nabla \sqrt{\pi}|^2$ and $\sqrt{\pi}$ vanishes on the boundary; the minimum is attained at $\pi = \phi_1^2$, with ϕ_1 the principal Dirichlet eigenfunction normalized in $L^2(\Omega)$. The LFP minimizes $I(p_\pi)$, so $d - B \le I(p_{\phi_1^2}) \le I(\phi_1^2) = 4\lambda_1(\Omega)$ at the LFP, and for large regions this bound is attained to leading order. In one dimension the leading term is Bickel's [1]. In general it is the second-order term of Levit [2], [3], in which the square of the ground state is the asymptotically least favorable prior [4, notes to Ch. 4]. We find that the next order is approximately a widening by the one-dimensional δ:

$$d - B \approx 4\lambda_1(\Omega_\delta), \qquad \text{(S2)}$$

on discs, ellipses, annuli (with the coupling of Section S1-D) and balls in $d = 3$, 4 and 5, up to an excess of the widening

SYMBOLS OF THE PAPER

The symbols of the main text, with the section of the paper where each is defined. Symbols local to one proof are defined at their point of use and are not listed.

Symbol	Meaning	Sec.
d, Z, φ_d	dimension; standard Gaussian noise in $\mathbb{R}^d$ and its density	II
Ω	compact region of the input	II
X, Y, π	input, output $Y = X + Z$, and the law of X	II
p_π	output density $\pi * \varphi_d$	II
$C(\Omega)$	capacity, in nats	II
$B(\pi)$	Bayes risk under squared loss; the LFP maximizes it	II
f_{data}, f_0, $G(\pi)$	data density of the NPMLE; $f_0 = 1/\lvert\Omega\rvert$ for uniform data, and the value $G(\pi)$ reported then	II
s	standard deviation of the Gaussian source of the NPMLE on curves and regions ($s = 2$)	II
δ_π	Bayes rule $\mathbb{E}_\pi[X \mid Y = y]$	II
D_π, r_π, J_π, F_π	optimality functions of the capacity, the LFP and the NPMLE; F_π when the problem does not matter	II
$I(p_\pi)$	Fisher information of the output density	II
Γ, $z(t)$	closed curve, parametrized by $t \in [0, 2\pi)$	II
$f(t)$, c_{2k}, M	formal density of order M on a curve and its cosine coefficients	II
A, m, R	half-width or radius of the region in the capacity, LFP and NPMLE problems	I
r_p, r_m	semi-axes of an ellipse; on the curves of Section VI, $z(0) = r_p$	I
h_d	entropy of the noise, $\frac{d}{2}\log 2\pi e$	III
β	edge constant of the capacity, $e^{C+h_1} = 2A + \beta$ up to exponentially small terms	III
L	half-length widened by the edge strip; $L = A + \beta/2$ for the capacity	III
K	number of atoms of an optimum	IV
u_j, w_j	positive atom positions and their masses on an interval	IV
K_b	number of atoms before an event	IV
N_{half}, ρ, v	atom count of the half-line solution within distance a of the wall, its density, and $v = 4\pi\rho$	IV
δ_b, e_k	correction of the births law, a series $e_2/v^4 + e_3/v^6 + \cdots$	IV
ϵ, λ, η	axis difference $r_p - r_m$, shape parameter and amplitude of the curves of (5); $\lambda = 0$ is the ellipse	VI
P1, P2	input confined to the filled region, or to the curve	VI
X pair, Y pair	atoms at the ends of the major axis ($t = 0, \pi$) and of the minor axis ($t = \pm\pi/2$)	VI
α_j	Verblunsky coefficients of the formal density	VI
τ	sign ± 1 of the para-orthogonal rule	VI
m_k, t_k, b_j	masses and angles of the atoms of a ring, and its harmonics	VI
c^*	reference level e^{-C-h_1} of the output density	III

that grows with d (Section S1-C). The limit of the excess is open.

Known upper bounds on capacity already have the form of a volume plus a boundary term [5], [6]; law (S1) replaces their constants by the measured edge strip. Several tests below draw on computations of the disc and ball NPMLE, of the response of capacity edges to curvature, and of the center events of Section S2; these computations are in the repository.

B. Tests

Table S1 lists the tests. In several of them the optimum had been computed before the prediction was registered and was read only afterwards; the status column marks each row. The LFP solvers covered discs and balls with radius $m \le 4.5$, ellipses with semi-axes up to 6 and annuli up to 5 wide, where the strip $\delta = 3.35$ is about as wide as the region; they did not converge at $m = 6.5$. At a center event the least favorable prior is a known radial law, with atoms at the center and on a few spheres. The event is the radius at which its optimality function first reaches the constant at a new point. Solving for that radius directly gives computed least favorable priors up to $m = 12.8$ in two dimensions, 13.2 in three, 13.1 in four and 12.1 in five. At the smaller sizes the departures from law (S2) are as large as the finite-size departures of the one-dimensional law itself (ratios 1.0139 at $m = 3.5$ and 1.0088 at $m = 4.5$), and a size-matched offset fits them equally well. On ellipses the bands did not separate the widened ellipse from the widened disc of equal area. The registered bands of the disc and ball tests are wider than the shift that the excess of Section S1-C causes in the ratio, so a pass in them does not separate δ from δ plus that excess.

C. The LFP strip in two to five dimensions

Read as a widening, the ball's effective strip $2j_{d/2-1,1}/\sqrt{d-B} - m$, with $j_{\nu,1}$ the first zero of the Bessel function J_ν, exceeds the one-dimensional δ by an amount that grows with d (Fig. S1). In two dimensions the excess rises to 0.0148 at $m = 12.8$ (0.44% of δ) and is still rising slowly. In three, four and five dimensions it reaches a maximum, 0.0385 near $m = 9.9$, 0.0635 near 7.9 and 0.0890 near 7.7, and then falls slowly, to 0.0367 at $m = 13.2$, 0.0583 at 13.1 and 0.0817 at 12.1. Whether it returns to δ is not established. Two registered forms for it, a curvature term in $1/m$ and a plateau, failed in $d = 3$, 4 and 5, and a registered test of its derivative in d at $d = 1$ failed (Section S7). The effective strip of the interval, $d = 1$, is $\pi/\sqrt{1-B} - m$, and it rises with m from 3.29252 at $m = 2.896$ to 3.33420 at $m = 9.064$ toward δ. Between two and three dimensions the difference between the effective strip in d dimensions and that of the interval at the same m scales as $d - 1$. The registered prediction for $d = 3$, the strip of the interval plus twice the two-dimensional difference, put the three-dimensional strips 3.38490 at $m = 7.589$ and 3.38584 at $m = 8.444$ above the values found by 0.00086 at both, inside the registered band of 0.003 (Table S1).

D. Close walls

When two walls are closer than the edge layers, the local law misses: for the NPMLE on annuli of width $W = 10$ to 18 the registered test failed. The one-dimensional problem explains

TABLE S1

TESTS OF THE VALUE LAWS (RATIO = OBSERVED/PREDICTED). STATUS: R, OPTIMUM COMPUTED AFTER REGISTRATION; E, OPTIMUM EXISTED BEFORE REGISTRATION AND WAS READ AFTER IT; S, INDEPENDENT RE-SOLVE OF A KNOWN STATE; P, READING AFTER THE FACT.

problem and region	result	status	log entry
capacity, disc ($A \geq 10$, 16 states): strip β	2×10^{-5} (three-term fit)	E	1510
capacity, superellipses $p = 2.5$, 3 ($A = 8$)	3.4×10^{-5}, 3.7×10^{-5} in C; constant term cancels	R	1513–1514
shape term	2.6%, 1.0% (resolution about 6%, 2.3%)	P	1521
capacity, d-balls, $d = 3, 4, 5$	strip $= \beta/2$	E	1516
capacity, curvature coefficient a_d from one dimension	$k_r = -0.63226$ (registered -0.633 ± 0.03)	E	1520
NPMLE, edge value on disc, annulus, ball	coefficient D_N, $2D_N$	S (R at $R = 33.5$)	1482–1489
NPMLE, four ellipses and two $\cos 4t$ stars	within bands; edge-average alternative excluded	R	1505, 1508
NPMLE, term of order one: 3D ball (coefficient a_3); 1D at a new width	$a_3 = -0.0413180$; width 20 to 5×10^{-7}	ball E and R; width R	1523, 1527
NPMLE, 3D shells	the ball's term of order one is a surface term	R	1541
LFP, discs $m = 2, 2.5, 3.5, 4, 4.5$ (band $[0.85, 1.15]$)	ratios 1.027, 1.017, 1.007, 1.003, 1.002	R	1558, 1564
LFP, ellipses $(5, 3.5)$, $(6, 4)$ (band $[0.97, 1.04]$)	ratios 1.005, 1.003	R	1560
LFP, balls $m = 2.5, 3.5, 4.5$ (band $[0.85, 1.15]$)	ratios 1.006, 0.999, 0.995	R	1566–1567
LFP, narrow annuli with the 1D coupling (band $[0.99, 1.03]$)	ratios 1.016, 1.008, 1.007 (uncoupled 1.073, 1.038, 1.033)	R	1565, 1568
LFP, d-balls at exact event states, $4 \leq m \leq 7.7$, $d = 2$ to 5	ratios 0.999–1.003, 0.993–0.995, 0.987–0.989, 0.981–0.984	E	1584–1585
LFP, widening in $d = 3, 4, 5$ at $9.5 \leq m \leq 13.5$ (two forms)	both forms fail in each d	R	1590, 1593, 1597, 1601, 1603, 1604
LFP, 3D excess from the 2D and 1D excesses, factor $d - 1$ ($m = 7.6$–8.4)	2 of 2 within 0.003 (misses -0.0009)	R	1591, 1602

The LFP bands were set after the one-dimensional values were known; the annulus band was set after the failure of the plain law on narrow annuli (entry 1563, Section S7). The d-ball bands were $[0.98, 1.02]$ for $d \leq 3$ and $[0.95, 1.05]$ for $d \geq 4$.

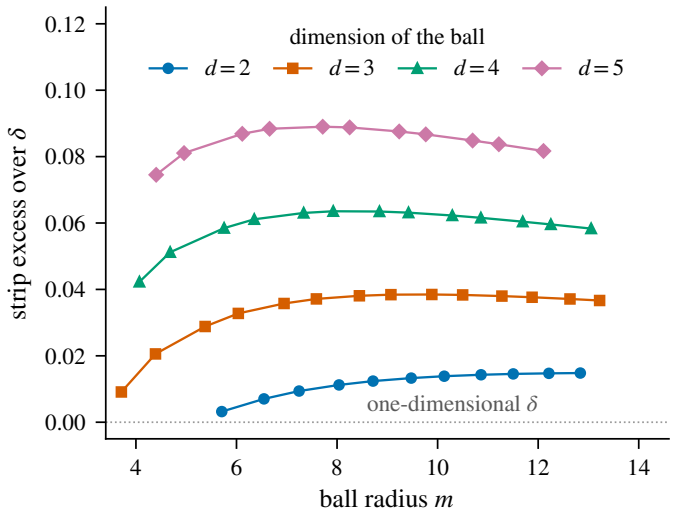


Fig. S1. The LFP strip of a d-ball is wider than the one-dimensional δ, rises to a maximum for $d = 3$, 4 and 5 and then falls slowly, and is still rising for $d = 2$; its limit is open. Excess of the effective strip over $\delta = 3.34776$ at the exact center-event states, for $d = 2$ to 5.

TABLE S2

CLOSE-WALL TEST OF THE NPMLE ON ANNULI WITH INNER RADIUS R_1 AND OUTER RADIUS R_2: OBSERVED MISS OF THE LOCAL LAW, STRIP TERM $4\pi R_{\text{mid}}\, \delta_0(W/2)$ AND THEIR RATIO.

annulus	miss	strip term	miss/term
$(12, 30)$	-2.5001×10^{-3}	-3.23956×10^{-3}	0.772
$(15, 30)$	-2.22299×10^{-2}	-2.43154×10^{-2}	0.914
$(20, 30)$	7.18304×10^{-2}	7.15058×10^{-2}	1.005
$(15, 25)$	5.77687×10^{-2}	5.72046×10^{-2}	1.010

TABLE S3

CLOSE-WALL TESTS OF THE LFP ON ANNULI WITH INNER RADIUS R_1 AND OUTER RADIUS R_2: RATIO OF THE OBSERVED DEFICIT TO THE PLAIN LAW $4\lambda_1(\Omega_\delta)$ AND TO THE COUPLED LAW $4\lambda_1(\Omega_\delta)\rho_{1D}(W/2)$. THE FIRST TWO ANNULI WERE TESTED WITHOUT THE COUPLING.

annulus	width	plain law	coupled law
$(6, 10)$	4	1.0491	—
$(8, 11)$	3	1.0695	—
$(7, 10)$	3	1.0733	1.0156
$(9, 13)$	4	1.0376	1.0081
$(10, 15)$	5	1.0327	1.0070

most of the miss. On an interval of half-width a the edges are already coupled, and the per-end defect of the NPMLE differs from D_N by an amount $\delta_0(a)$; for capacity, e^{C+h_1} differs from $2a + \beta$ by $\varepsilon(a)$. Taken at half the width and carried along the midline of radius R_{mid}, these deviations predict the misses on annuli.

For the NPMLE the observed miss is 77 to 101% of $4\pi R_{\text{mid}}\, \delta_0(W/2)$ on the four annuli of Table S2. For capacity the coupling $2\pi R_{\text{mid}}\, \varepsilon(W/2)$ passes at 19 of 19 decisive entries on narrow annuli (13 distinct widths), and $4\pi R_{\text{mid}}^2 \varepsilon(W/2)$ at 12 of 14 decisive entries on a 3D shell (10 distinct widths); on eccentric annuli the coupling follows the local width. For the LFP the coupling is a factor, the one-dimensional finite-width ratio $\rho_{1D}(W/2)$. The two annuli of the first test, narrower than 2δ, exceeded the plain law by 5% and 7% (Table S3), and $4\lambda_1(\Omega_\delta)\rho_{1D}(W/2)$ holds on three further annuli to 0.7 to 1.6% (Table S3). What remains after the coupling is up to 14% (2D) and 33% (3D) of it for capacity and up to 23% of the term for the NPMLE, where it is positive on all four annuli of Table S2. It is not explained.

S2. Center events in d dimensions

Status. The shift rule (S3) is an empirical two-point rule that was tested blind in higher dimensions. The radius $A = \sqrt{d+2}$ of the first NPMLE event is the zero of the exact second derivative $A^2/(d+2) - 1$ of the single center atom. The other statements of this section are numerical.

On a d-ball the center alternates between a birth, where an atom appears at the center, and a lift-off, where the center atom becomes a small sphere; on an interval the sequence is birth and split (Section III). To pair the events, count a center atom as one atom and a sphere as two, matching the pair of atoms at $\pm r$ on the interval; the one-dimensional partner of an event on the ball is the interval event with the same count. The first LFP events are the classical thresholds: 1.0567 on the interval [7] and the threshold of the uniform prior on the sphere [8], [9], tabulated in Table I of [10]. The shift Δ_d of each event n from its partner follows, to first order,

$$\Delta_d = (d-1)(u_n + v_n d), \tag{S3}$$

with the two parameters u_n and v_n of event n fixed by $d = 2$ and 3. The form was suggested by McMahon's expansion of the zeros of Bessel functions, $j_{\nu,s} \approx (s + \nu/2 - 1/4)\pi - (4\nu^2 - 1)/(8(s + \nu/2 - 1/4)\pi)$ with $\nu = d/2 - 1$ before a birth and $\nu = d/2$ before a lift-off. The numerator $4\nu^2 - 1$ equals $(d-1)(d-3)$ for a birth and $(d-1)(d+1)$ for a lift-off, so it is $d-1$ times a linear function of d. Fitted event by event, however, the two parameters do not behave as one local wavenumber and one radius, so we use (S3) as an empirical two-point rule in d. The blind predictions in higher dimensions therefore test the smoothness of $\Delta_d/(d-1)$ in d, and not the Bessel mechanism that suggested the form.

Registered blind, the rule matched the LFP events e_3 to e_7 in $d = 4$ and 5 within 0.64% and 1.81% (Table S4), and the NPMLE events in $d = 4$ within 1.31%, 16 of 17 within 1%.

On capacity events computed earlier, registered but not blind, it held within 1% in $d = 4$ and 2.8% in $d = 5$. The first LFP events miss in $d = 4$ and 5, by up to 12.7%.

In $d = 8$ the first events of every problem miss, by 18.6% for the NPMLE at $K_{\text{eff}} = 2$ and 13.4% for capacity at $K_{\text{eff}} = 4$ (read after the runs), and the registered tests there failed. The failed tests are the shared residual of the NPMLE and capacity (10 of 13 events within 0.5%, 11 needed) and the LFP events e_3 and e_4 ($+8.5\%$ and $+5.8\%$ against a band of 5%; e_5 to e_7 lie inside; Section S7). The first LFP event in $d = 8$, which is not among the scored events, misses by 88%: its shift is 2.138 against 1.137 from (S3). A simpler form, linear in $d-1$, also stays inside the bands at many later events, with misfits up to ten times those of (S3); the early events are where the bands discriminate.

The NPMLE events have a second reading. With uniform data on a ball the NPMLE is a rate–distortion problem, and its center events are the critical temperatures of deterministic annealing for a uniform ball source. The first lies where the variance of the source along an axis equals the noise variance. For a uniform ball of radius A that variance is $A^2/(d+2)$, so the first event sits at $A = \sqrt{d+2}$. At that radius the critical temperature of deterministic annealing, twice the largest eigenvalue of the source covariance [11], [12], equals the temperature $T = 2$ that corresponds to unit noise variance. The instability of the single center atom there is exact (its second derivative is $A^2/(d+2) - 1$), and the computed first event sits at this radius in $d = 1$, 2, 3, 4 and 8. In $d = 2$, 3, 4 and 8 the radius was registered in advance of the run. The blind NPMLE test in $d = 4$ thus predicted a sequence of critical temperatures of deterministic annealing from the events in $d = 2$ and 3.

S3. The continuation past the birth

Status. The statements of this section are numerical comparisons of computed optima with the formal construction. None is proved.

For capacity the construction continues past the birth. We write the input as $(1 - m_0)$ times a formal density on the curve plus m_0 times an atom at the center, and ask that the curve harmonics vanish and the center reach C. At $r_p = 2.8$, $r_m = 2.200$ this predicted the mass $m_0 = 0.0014575498$, and a ring with a center atom gave the same value to 10^{-10}. The Hessian at the center tells when the center atom becomes unstable. On the disc this happens at radius 3.42035, inside the bracket 3.402 to 3.460 that we obtained from the vector data of Fig. 5(a) of [13], on the grid points of the figure and with our conversion $R = \sqrt{2\,\text{pSNR}}$ (a check made after the fact). On the ellipse the major-axis direction goes first: at $r_p = 3.5$ we registered the split at $r_m = 3.20334$ with a band of $\pm 2 \times 10^{-4}$, and a 40-atom ring with a center atom changes the sign of that eigenvalue at 3.203334, 6×10^{-6} below the registered value. Past it the center atom becomes a pair on the major axis, in a supercritical pitchfork; the half-separation of the pair, registered at $r_m = 3.22$ as 0.097617, was observed with every printed digit.

S4. Rings inside rings

Status. The statements of this section are numerical tests of the copying rule on inner rings. None is proved.

The inner pair is itself a ring, on a curve that nothing prescribes. We describe the inner part by a formal density of its own, of mass m_i, on a free curve $r(\theta) = \sum_\ell r_\ell \cos 2\ell\theta$. It is fixed together with the boundary density by asking that the optimality function and its radial derivative be flat on the inner curve to the same order. On an ellipse the copying rule is a statement about complex moments, and on the inner curve we read it in the curve's own eccentric parameter u, the angle in $(a\cos u, b\sin u)$ for an inner ellipse of semi-axes a and b. The pair is then lost where the second harmonic of the inner density in u reaches 2, the value of a pair at $u = 0, \pi$ and the largest that a probability law allows; the four-atom ring that follows is lost where the fourth harmonic does. We call these two losses a pair (two atoms of the inner ring) and a quartet (four). Table S5 summarizes the registered tests, and Table S6 lists each of them with its band, the observed loss and whether it fell inside the band. The rule passed in all three problems at small inner eccentricity ϵ_i, on ellipses and on the mixed curves. Each problem failed one registered test, at the most elongated inner ring we tried: the capacity pair at $\epsilon_i \approx 0.13$,

TABLE S4
SHIFT RULE (S3) FOR THE LFP CENTER EVENTS e_3 TO e_7. THE SHIFTS s_2 AND s_3 OF AN EVENT FROM ITS ONE-DIMENSIONAL PARTNER IN $d = 2$ AND 3 FIX $u_n = 3s_2 - s_3$ AND $v_n = s_3/2 - s_2$, SINCE $s_2 = u_n + 2v_n$ AND $s_3 = 2(u_n + 3v_n)$; THE SHIFTS IN $d = 4$ AND 5 WERE REGISTERED BLIND, WITH PREDICTIONS $3(s_3 - s_2)$ AND $6s_3 - 8s_2$ COMPUTED FROM THE UNROUNDED SHIFTS, SO THEY CAN DIFFER FROM THE PRINTED DIGITS IN THE FIFTH DECIMAL. ERROR IS OBSERVED OVER PREDICTED MINUS ONE, IN PERCENT.

event	s_2	s_3	u_n	v_n	$d=4$: observed	predicted	error	$d=5$: observed	predicted	error
e_3 birth	0.42601	0.81470	0.4633	−0.0187	1.17354	1.16608	+0.64	1.50698	1.48016	+1.81
e_4 lift-off	0.33146	0.64096	0.3534	−0.0110	0.93244	0.92848	+0.43	1.20880	1.19404	+1.24
e_5 birth	0.41268	0.80712	0.4309	−0.0091	1.18468	1.18330	+0.12	1.54581	1.54122	+0.30
e_6 lift-off	0.34325	0.67249	0.3573	−0.0070	0.98910	0.98772	+0.14	1.29435	1.28895	+0.42
e_7 birth	0.40920	0.80878	0.4188	−0.0048	1.19854	1.19873	−0.02	1.57775	1.57906	−0.08

TABLE S5
REGISTERED LOSSES OF INNER RINGS (THE FULL LIST IS TABLE S6). COUNTS ARE PASSED/REGISTERED; ϵ_i IS THE LARGEST INNER ECCENTRICITY TESTED.

problem	pairs	quartets	failure
capacity	3/4	4/4[a]	pair at $\epsilon_i \approx 0.13$
LFP	2/3	2/2[a]	pair at $\epsilon_i = 0.048$
NPMLE ($s = 2$)	3/4	2/2[a]	pair at $\epsilon_i = 0.10$

[a] Two of the four capacity quartets (at $r_p = 4.3$ and on the mixed curve), the two LFP quartets and the two NPMLE quartets were registered with a band that contains a fitted or set second-order term. The other two capacity quartets used exact complex-moment bands. Six further capacity events, the pair and the quartet at $r_p = 3.7$, 3.85 and 4.0, passed offset bands registered in polar form.

the LFP pair at 0.048 and the NPMLE pair at 0.10, each lost below the predicted point by 0.4 to 0.9% in the harmonic. The misses grow with ϵ_i, and the capacity quartet passed at $\epsilon_i \approx 0.13$ only because its band included a fitted second-order term. The capacity quartet on the mixed curve and the four quartets of the other two problems were registered with bands that also contain a second-order term (Table S5), so the quartet counts test the first order only. So the inner rings follow the rule to first order in their eccentricity, and the second-order term differs between the problems.

S5. WHAT THE SERIES LEAVES: A RESIDUAL TIED TO THE AMBIGUITY OF THE RESUMMATION

Status. The residual and its relation to the ambiguity of the resummation are numerical findings. Nothing in this section is proved, and the cause of the residual is not derived.

a) Setting: Let p be the output density, the density of the Gaussian mixture of the atoms, let φ be the Gaussian kernel and let W be a uniform weight on a window. The free-edge family with exponent k has the optimality function $\int \varphi\, W p^{-k}$, the integral of the Gaussian kernel centered at the point against Wp^{-k}, and all its atoms are free. The NPMLE is its member $k = 1$, and the immune problem is its member $k = 2$. The hard-wall family has the function $(p^{-k} - 1)/k$ in place of Wp^{-k}, with the atoms confined to $[-A, A]$ and one atom fixed at the bound. For $k \to 0$ this function tends to $-\log p$, and the member $k = 0$ is the capacity problem.

Below the tower, at $K_b < 60$, the density ρ is lower and the series is less accurate. The law with the series to order 16 still gives each of the 47 stored capacity amplitudes in $K_b = 10$ to 59 within a bar of 0.3×10^{-9} plus three times the change between the orders 14 and 16. Three of the 50 values of K_b in this range have no stored event. The largest deviation is 0.39×10^{-9} for $K_b = 40$ to 59 and 5.8×10^{-9} for $K_b = 24$ to 39. These are readings of events that were known. For the NPMLE, 17 stored transitions in $K_b = 31$ to 52, which holds 22 values, lie inside the bar, and all 23 deviations at $K_b = 6$ to 28 are positive.

The series is divergent, so a sum has to be chosen. We take the median of its two lateral Borel sums, computed from a diagonal Padé approximant of the Borel transform. This leaves a small positive residual R in the amplitude. It is not noise: it tracks the *ambiguity* of the resummation, which is the imaginary part of one lateral sum, converted to an amplitude and written $D_a(v^2)$, where $v = 4\pi\rho$ and ρ is the density of atoms (Section IV-B of the paper).

For each functional the ratio $\sigma = R/D_a$ changes by a few percent while R itself falls by a factor between 280 and 13000. A common factor $1 - 14.6/v^2$ describes the remaining change to 0.4% in $\ln R$ over 54 events with $K_b \le 40$ and $K_b \neq 38$, namely 16 for $k = 1$, 26 for $k = 2$ and 12 for $k = 3$.

The multipliers are 3.09, 5.55 and 7.90 for the NPMLE problem ($k = 1$), the immune problem ($k = 2$) and a third functional ($k = 3$).

b) Tests: The multiplier grows with k. At $k = 3$ it was predicted before the ten events $K_b = 28$ to 37 were computed: the registered interval for the ratio of its residual to that of the immune problem was 1.35 to 1.55, and the measured ratio was 1.414 to 1.424. At $k = 4$ two continuations were registered, a line (1.308) and a quadratic through $k = 1, 2, 3$ (1.278). The measured ratio is 1.2865, so the line is rejected and the quadratic lies inside its registered band of 0.9%. For $k = 1.5$, 2.5 and 5 the amplitudes were predicted to 0.1% before they were computed. A five-parameter description of 243 events of ten exponents, $k = 0.75$ to 12, including the events of the failed test described below, was then tested blind at $k = 6$, 10 and 16 and passed its bars of 5%, 5% and 15%, with an overprediction of up to 8% at $k = 16$, four units beyond the fitted range.

c) What failed: Before that description, straight lines in k for the parameters of the fit were fixed in advance and tested blind at $k = 0.75$, 8 and 12. At $k = 0.75$ the measured residual was within 1.7% of the prediction at 18 of the 19 events and within 1.9% at the other event, inside the registered bar of 6%. At $k = 8$ the ratio of measured to predicted residual ran from 1.07 to 1.14, and at $k = 12$ from 1.19 to 1.43, outside the registered bars of 10% and 15%. The test failed by the

TABLE S6

EVERY REGISTERED LOSS OF AN INNER RING (SECTION S4; SUMMARY IN TABLE S5). FORM: CM, COMPLEX MOMENT; E, ECCENTRIC; P, POLAR WITH AN OFFSET BAND. E^a: ECCENTRIC POINT PLUS A RESIDUAL FITTED ON CAPACITY (0.15 TO $0.35\,\epsilon_i^2$; LFP $0.25\,\epsilon_i^2 \pm 5\times10^{-4}$); E^b: ECCENTRIC POINT -5×10^{-4} TO $+0.25\,\epsilon_i^2 + 5\times10^{-4}$; E^c: ALSO REGISTERED WITH A MISS SPECIFIC TO THE PROBLEM OF $-2.0\times10^{-4} \pm 5\times10^{-5}$ (OBSERVED -1.95×10^{-4}). NPMLE: GAUSSIAN SOURCE, $s = 2$. OBSERVED VALUES ARE ZEROS BETWEEN COMPUTED POINTS.

problem, region, r_p	ring, form	registered	observed	inside the band
capacity, ellipse, 3.6	pair, CM	3.39615–3.39695	3.39656	yes
capacity, ellipse, 3.6	quartet, CM	3.43054–3.43134	3.43094	yes
capacity, ellipse, 4.15	pair, CM	3.31403–3.32113	3.31902	yes
capacity, ellipse, 4.15	quartet, CM	3.50003–3.50386	3.50338	yes
capacity, ellipse, 4.3	pair, E	3.29308–3.29428	3.29254	no
capacity, ellipse, 4.3	quartet, E^a	3.52424–3.52616	3.52568	yes
capacity, mixed 0.2, 3.7	pair, E	3.38143–3.38243	3.381956	yes
capacity, mixed 0.2, 3.7	quartet, E^b	3.44158–3.44269	3.442136	yes
LFP, ellipse, 2.6	pair, E	2.27253–2.27333	2.272753	yes
LFP, ellipse, 2.6	quartet, E^a	2.32039–2.32139	2.320752	yes
LFP, mixed 0.1, 2.6	pair, E	2.27134–2.27234	2.271647	yes
LFP, mixed 0.1, 2.6	quartet, E^b	2.32083–2.32192	2.321326	yes
LFP, ellipse, 2.8	pair, E	2.24645–2.24745	2.246310	no
NPMLE, ellipse, 2.9	pair, E	2.53389–2.53489	2.534208	yes
NPMLE, ellipse, 2.9	quartet, E^b	2.60478–2.60621	2.605352	yes
NPMLE, mixed 0.2, 2.9	pair, E	2.53134–2.53234	2.531636	yes
NPMLE, mixed 0.2, 2.9	quartet, E^b	2.60374–2.60517	2.604292	yes
NPMLE, mixed 0.1, 2.9	pair, E^c	2.53262–2.53362	2.532921	yes
NPMLE, ellipse, 3.1	pair, E	2.49899–2.49999	2.498436	no
capacity, ellipse, 3.7–4.0	6 events, P	offset bands	inside	yes

letter (Section S7). The capacity residual does not continue the sequence in k: its ratio R/D_a rises from 0.032 to 0.64 over $K_b = 20$ to 45, whereas the free-edge family extrapolated to $k = 0$ gives 0.10 to 0.18 at $K_b = 29$ to 37, where the capacity gives 0.34 to 0.52. The capacity problem has an atom fixed on the amplitude bound, and the free-edge problems do not. A family with that hard wall contains the capacity at $k = 0$. A three-term series in $1/v^2$ with nine fitted constants describes it to an rms of 0.00087 in R/D_a over fourteen exponents. It predicted three new exponents $k = 0.15$, 0.4 and 0.6 within 0.0035 against a bar of 0.006. None of the constants is derived.

The cause of the residual is not derived. We read it as the real part of a term tied to the ambiguity of the resummation, with one parameter for each problem.

S6. THE STRUCTURE OF THE COEFFICIENTS OF THE BIRTHS LAW

Status. The values of the coefficients are computed exactly, and the patterns found in them are observations. None of the patterns is proved.

The finer version of the births law (Section IV-B) takes the density of the atoms from a series, the bulk law, whose coefficients F_j are polynomials in $u = 1/\pi^2$ with rational coefficients. We computed them exactly through F_{15}, and the center coefficients e_n exactly through e_{10}. The center coefficients e_{11} to e_{20} are stored as numerical values. On these values we found a structure. The coefficient of u^j in F_j is exactly 8^{-j} for $j = 2$ to 15. The coefficient of u^{j-1} times 8^j equals $-j(j-1)(6j^3 + j^2 - 19j + 16)/90$. We found this polynomial on $j = 2$ to 10 after the fact, and it then held exactly on $j = 11$ to 15, which were not used to find it. The logarithm of the generating function of the F_j has the coefficient $1/(n\,8^n)$ at u^n for $n = 1$ to 15. Nothing here is proved, and a fit to stored data can always be adjusted. A test that fitting cannot satisfy needs a coefficient that does not exist yet; predictions for e_{11} and F_{16} are registered in advance in the registration log. They have not been scored, and none of their results is used here.

S7. WHAT DOES NOT CARRY OVER, AND THE TESTS THAT FAILED

Status. The ring counts and the midpoint rule are numerical. Table S7 lists the registered tests whose outcome was not a pass. The larger conjecture below is a conjecture, and the blind test described there is its only test.

Some structure differs between the problems. Counting a center atom as one ring, the capacity disc ($A = 7$ to 11) and the NPMLE disc ($R = 10$ to 25) have $\lfloor K_{1D}/2 \rfloor$ rings, where K_{1D} is the atom count on $[-A, A]$. This follows from the pairing of events in Section S2 away from the shifted events. The number $\lfloor K_{1D}/2 \rfloor$ is the number of symmetric pairs of atoms of the one-dimensional optimum. The center atom of the disc need not be the center atom of the interval, so the rule counts rings and says nothing about layouts. The LFP disc has one ring more at $m = 2$ and 3.5, and the registered ring-count test failed at $m = 4$. Its center atoms are born at $m = 1.53499$, 3.32183 and 4.98162, near the midpoints of consecutive one-dimensional events (1.05674 and 2.00365; 2.89582 and 3.74808; 4.56894 and 5.36404). So the disc gains a ring about half an event spacing early. The midpoint rule missed by 0.0048 (after the fact), 0.00012 (registered) and 0.015 (registered, outside its band of 0.01).

Supports also differ where values agree. On superellipses with $p > 2$, solved at $A = 3$, the capacity wall support breaks into pieces, while the value law holds at $A = 8$. Two registered support readouts failed: the sector counts of the eccentric NPMLE support at two eccentricities, on sampling gaps, and the effective radius read from the LFP output density.

Open problems of the planar results. The copying and loss conjectures lack a proof, and the linear-response argument is perturbative. The inner rings follow the rule only to first order, and the second-order term differs between the problems. The limit of the LFP strip in $d \geq 2$ is unknown. The closed-curve rule and the one-dimensional count law are two mechanisms that meet only at the first two scalar changes (Section VI); a single derivation that contains both is not yet known.

The larger conjecture. The following conjecture is not established in this paper. Whenever a law on a bounded region is chosen optimally and seen through Gaussian noise, its atoms follow one description, with the count, the positions, the edge layers and the births fixed by the same one-dimensional constants. Other problems have this form exactly. A prior that maximizes the information a model's data carry about its parameters is a capacity-achieving input, and Mattingly et al. found the $4/3$ exponent there [14]; we have made no separate test there. A decision made under squared-error loss and a cost on mutual information is a rate–distortion problem [12]. We ran a blind test against twelve published one-dimensional solutions of such problems, from rate–distortion and from rational inattention; the file bridge_specs.md of the repository lists the cases with their sources. Our NPMLE solver predicted the number of reproduction points or actions in nine of the twelve: eight of nine with a uniform source and one of three with a truncated normal one. The next steps are a proof of the count law, the limit of the LFP strip, and registered tests of the planar description in these problems. If the description holds there, an optimal constellation, prior, quantizer or decision rule could be read off a smooth density in each of them. If it fails, the failure will show which part of the mechanism is specific to the three problems treated here.

The first births law. The first version of the births law of Section IV-B of the paper cut the series at e_5 and took the density from a fit to the half-line atoms. Two of its registered tests failed by the letter: the step of its residual, registered to lie in 1.4 to 2.6, was 2.68 at $K_b = 71$, and the ratio of consecutive residuals, registered in 1.08 to 1.19, was 1.1917 at $K_b = 78$. The residual of this version, from 2.4 at $K_b = 61$ to 16.1 at $K_b = 80$, is the part of the later terms that the density does not cancel. The law is also sensitive to two instrument choices: the half-line solution must be converged in its depth, and a cubic spline of N_{half} adds an error of up to 0.020 at births, which the quintic spline removes.

Table S7 lists the registered tests whose outcome was not a pass; runs that were not accepted, were stopped or were invalid are listed in the repository.

TABLE S7
REGISTERED TESTS WHOSE OUTCOME WAS NOT A PASS: FAILURES, TESTS THAT COULD NOT BE DECIDED, THIRD OUTCOMES AND SOLVER FAILURES

outcome	log entry
failed: form S on the ellipse $r_p = 3$, 18-ring ($+1.18\%$)	1356
failed: extremal form, mixed curve, 18-ring (-2.81%)	1331
failed: inner pairs at the largest ϵ_i (three problems)	1438–1477
failed: NPMLE local law on narrow annuli	1530
failed: the tilt dependence of the one-dimensional finite-width deviation as the explanation of the capacity remainder	1539
failed: NPMLE sector counts of the eccentric support at $e = 2.5$, 3, on sampling gaps	1548, 1550
failed: plain LFP law on annuli narrower than 2δ	1563
failed: LFP ring count at $m = 4$ (as registered)	1564
failed: LFP bulk shape	1571
failed: first LFP center events of the 3D ball, predicted from the one-dimensional sequence	1573
failed: 2D birth e_5 at the 1D midpoint (0.015, band 0.01)	1574
failed: second LFP lift-off in 2D and 3D at the same fraction of its one-dimensional interval as the first	1577
failed: shared residual, NPMLE and capacity, $d = 8$	1587
failed: LFP center events e_3, e_4 in $d = 8$	1589
failed: LFP widening forms in $d = 3, 4, 5$	1601, 1603, 1604
failed: derivative of the LFP widening in d at $d = 1$	1599
failed: first births law, residual step at $K_b = 71$	1820
failed: first births law, residual ratio at $K_b = 78$	1916
missed: table of the first variant, NPMLE transition $K_b = 93$	2040
failed: table of the first variant, capacity birth $K_b = 94$	2160
failed: residual closure linear in k (blind, $k = 0.75$, 8, 12)	2075
third outcome, just outside a registered interval: the bias of $D'(0)$, and the event of a curved strip, which came earlier than in one dimension	1525, 1552
not decidable: NPMLE at $R = 45$ (bulk structure below the float64 floor); $D'(0)$ at $a = 10$ (the two edges interact); LFP ring count (both accepted states as predicted)	1484, 1529, 1558
solver failed validation: the 2D ellipse solver for the NPMLE failed its validation on the disc	1491
result: the small-tilt edge response has two different constants, one for the NPMLE and one for capacity	1503

S8. PROOF OF THE DEFORMATION IDENTITY OF THE NPMLE

Status. Section S1 uses an exact identity for the derivative of the NPMLE value when the region is deformed. This section proves it for the unconstrained problem defined below. The local law that follows from it is not proved, and it rests on two stated assumptions at the end of this section.

a) Setting: Let $\Omega \subset \mathbb{R}^d$ be a bounded open set with C^1 boundary, $f_0 = 1/|\Omega|$, and φ the standard normal density. For a probability measure G on $\mathbb{R}^d$ put $m_G = \varphi * G$ and

$$L(\Omega) = \sup_G f_0 \int_\Omega \log m_G \, dx,$$
$$\widetilde{G}(\Omega) = \int_\Omega \log(m^*/f_0)\, dx = \frac{L(\Omega) - \log f_0}{f_0},$$

where m^* is the optimal output density. The supremum is over all probability measures on $\mathbb{R}^d$. If Ω is convex, the argument of part (a) with $K = \overline{\Omega}$ shows that every maximizer is supported in $\overline{\Omega}$, so this is the problem of Section II of the paper. For a non-convex Ω, such as an annulus, the problem of that section confines G to Ω and differs from this one, and the proposition does not cover it. A deformation is $\Omega_\epsilon = \Phi_\epsilon(\Omega)$ with $\Phi_\epsilon(x) = x + \epsilon V(x)$, where $V \in C^1_c(\mathbb{R}^d; \mathbb{R}^d)$ and $|\epsilon|\, \|DV\|_\infty < 1$, where $\|DV\|_\infty = \sup_x \|DV(x)\|_{\mathrm{op}}$ is the supremum of the operator norm. Then Φ_ϵ is a C^1 diffeomorphism: it is injective because $|\Phi_\epsilon(x) - \Phi_\epsilon(y)| \geq (1 - |\epsilon|\|DV\|_\infty)|x - y|$, by the mean value inequality for V. It is a local diffeomorphism with $\det(I + \epsilon DV) > 0$: the matrix $I + \epsilon DV(x)$ is invertible because

$|\epsilon|\,\|DV(x)\|_{\mathrm{op}} < 1$, so the determinant never vanishes, and it is continuous in ϵ and equals 1 at 0. It is surjective: it is open, being a local diffeomorphism, and its image is closed, because $\Phi_\epsilon - \mathrm{id}$ is bounded and so Φ_ϵ is proper, and $\mathbb{R}^d$ is connected. Hence Ω_ϵ is bounded and its boundary $\Phi_\epsilon(\partial\Omega)$ is C^1. The normal speed is $\delta_n = V \cdot \nu$ on $\partial\Omega$, and $|\Omega_\epsilon| = \int_\Omega \det(I+\epsilon DV)\,dx$ has the derivative $\int_\Omega \operatorname{div} V\,dx = \oint \delta_n\,ds$ at 0, by the divergence theorem on a bounded domain with C^1 boundary.

Proposition S1. *(a) A maximizer exists, every maximizer is supported in any closed ball K that contains $\overline{\Omega}$, and m^* is unique on $\mathbb{R}^d$. (b) $L(\Omega_\epsilon)$ is differentiable at 0 and* $\dfrac{dL}{d\epsilon} = f_0 \oint \delta_n\,(\log m^* - L)\,ds$. *(c) $\widetilde{G}(\Omega_\epsilon)$ is differentiable at 0 and*

$$\frac{d\widetilde{G}}{d\epsilon} = \oint \delta_n \Big[\log \frac{m^*(s)}{f_0} + 1\Big] ds. \tag{S4}$$

In words: the shape derivative of $\widetilde{G}$ at a boundary point equals $\log(m^*/f_0)+1$ there. For a disc of radius R and $\delta_n = 1$ this reads $\log(m^*(R)/f_0) = -1 + \widetilde{G}'(R)/(2\pi R)$.

Proof. (a) For θ outside K and $x \in K$, the projection Π_K onto K satisfies $|x-\Pi_K\theta|^2 \le |x-\theta|^2 - |\theta-\Pi_K\theta|^2$. So moving the mass outside K to its projection strictly raises $m_G(x)$ at every $x \in \Omega$, and every maximizer lives on K. The set $\mathcal{P}(K)$ of probability measures on K is compact and metrizable in the weak topology. For $x \in \Omega$ we have $m_G(x) \ge c > 0$, and $G \mapsto m_G(x)$ is weakly continuous, so the objective is weakly continuous on $\mathcal{P}(K)$ by dominated convergence, and a maximum exists. The objective is concave in G and strictly concave in $m_G|_\Omega$. Hence two maximizers have equal outputs on Ω, and since m_G is entire for compactly supported G, they agree on $\mathbb{R}^d$.

(b) Choose K to contain $\overline{\Omega_\epsilon}$ for $|\epsilon| \le \epsilon_1$, and put $F(\epsilon, G) = |\Omega_\epsilon|^{-1}\int_\Omega \log m_G(x+\epsilon V)\det(I+\epsilon DV)\,dx$. Then $L(\Omega_\epsilon) = \max_{G\in\mathcal{P}(K)} F(\epsilon, G)$. The function F is C^1 in ϵ, and $\partial_\epsilon F$ is jointly continuous on $(-\epsilon_1,\epsilon_1)\times\mathcal{P}(K)$, since m_G and ∇m_G are weakly continuous in G and equi-Lipschitz in x, and $m_G \ge c$. A weakly convergent sequence $G_n \to G$ then has $m_{G_n} \to m_G$ and $\nabla m_{G_n} \to \nabla m_G$ uniformly on compact sets, by the Arzelà–Ascoli theorem. Danskin's theorem [15]–[17] then gives one-sided derivatives equal to the maximum and the minimum of $\partial_\epsilon F(0, G)$ over the maximizers. In detail, $f(\epsilon) = \max_G F(\epsilon, G)$ is continuous, being the maximum over a compact set of a jointly continuous function, and satisfies $f(\epsilon) \ge F(\epsilon, G^*)$ for each maximizer G^* at 0. For the converse take $\epsilon_n \downarrow 0$ and maximizers G_n; the mean value theorem gives $f(\epsilon_n) - f(0) \le \epsilon_n\,\partial_\epsilon F(\xi_n, G_n)$ for some $\xi_n \in (0, \epsilon_n)$, and a weakly convergent subsequence $G_n \to G$ has $F(0,G) = f(0)$ by continuity of f. By the divergence theorem, $\partial_\epsilon F(0,G) = |\Omega|^{-1}[\oint \delta_n \log m_G\,ds - F(0,G)\oint \delta_n\,ds]$, which depends on G only through m_G, and $m_G = m^*$ for every maximizer by (a). So the right derivative is this value. For $\epsilon_n \uparrow 0$ the same argument with the inequalities reversed gives the left derivative, with the same value. This gives (b).

(c) Put $P = \oint \delta_n\,ds = |\Omega|'$. Then $f_0' = -f_0^2 P$ and $L = \log f_0 + f_0\widetilde{G}$ give $L' = -f_0 P - f_0^2 P\widetilde{G} + f_0\widetilde{G}'$. By $\log m^* - L = \log(m^*/f_0) - f_0\widetilde{G}$, part (b) reads $L' = f_0 \oint \delta_n \log(m^*/f_0)\,ds - f_0^2\widetilde{G}P$. Equate the two expressions for L'. The terms $f_0^2\widetilde{G}P$ cancel, and division by f_0 leaves (S4), because $P = \oint \delta_n\,ds$. □

b) From the identity to the local law: Two assumptions, each of which can fail, turn (S4) into a law for the edge value. (A1) *Locality with a differentiable remainder.* For convex planar regions with all length scales at least R we have $\widetilde{G}(\Omega) = \oint \big(D_N + a\kappa + b\kappa^2 + c_3\kappa^3 + \cdots\big)ds + \rho_N(\Omega)$ with constants that depend only on the noise, where the remainder ρ_N is C^1 in the shape and its shape derivative has the form $\rho_N'(\Omega)[\eta] = \oint r_N \eta\,ds$ with $\sup|r_N| = O(R^{-n})$ for every n. A remainder that is only small in value is not enough, because its derivative could be of order one. (A2) *A flat edge.* The half-plane optimum is the one-dimensional half-line optimum times the uniform law along the edge, and D_N is the one-dimensional edge defect. For a closed convex curve with outward normal, the first variations under a normal speed η are $\delta(ds) = \kappa\eta\,ds$ and $\delta\kappa = -(\eta'' + \kappa^2\eta)$. The variation of $\oint \kappa\,ds$ is 0 by Gauss–Bonnet, and

$$\delta \oint b\kappa^2\,ds = b\oint \big(2\kappa\,\delta\kappa + \kappa^3\eta\big)ds = b\oint \eta\,(-2\kappa'' - \kappa^3)\,ds,$$

after two integrations by parts. Since (S4) holds for all η, we get the formal expansion $\log(m^*/f_0) + 1 = D_N\kappa - b(\kappa^3 + 2\kappa'') + \cdots$, the local law of Section S1. Numerically, the two sides of (S4) were compared on discs for a uniform dilation, $\delta_n = 1$. The left side is $\widetilde{G}'(R)/(2\pi R)$, with the derivative of $\widetilde{G}$ taken by a centered difference in R of step 0.05 on the computed optima, and the right side is $\log(m^*(R)/f_0) + 1$. They agree to 1.6×10^{-5} at $R = 10$ and to 3×10^{-7} at $R = 20$, relative to the right side. This check does not test the terms with η''. With the disc fit $\widetilde{G} \approx 2\pi D_N R + 2\pi a + 2\pi b/R$, the truncation error of the centered difference is about $h^2\widetilde{G}'''/6$ with $\widetilde{G}''' = -12\pi b/R^4$ and h the step, which is more than two orders of magnitude below these gaps at both radii. What the gaps consist of is not established. A proof of (A1) would need the decay of the bulk ripple faster than any power, the decoupling of the edge layers, and a control of the remainder and of its shape derivative.

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